From Friend to Partner: Designing Consent-Aware Relationship Transitions in AI Companions

Design Component Report

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Human-Computer Interaction

I. Research Questions

Our project studies how an AI companion should handle a change from a friendly relationship toward a romantic one. We continue to address the following research questions from our proposal:

1) How do users feel when an AI companion becomes romantic without asking first?
2) Which approach do users prefer: automatic change, asking permission first, or a user-controlled relationship setting?
3) What helps users feel comfortable, in control, and able to trust the AI during this change?

The design component turns our formative findings into a digital prototype. We first mapped the affinity-diagram findings to design takeaways, then defined user tasks, explored paper sketches, conducted peer critique, and finally developed a higher-fidelity prototype named *Aiva*.

II. From Affinity Diagram to Design Takeaways

Our interview responses were organized into seven affinity-diagram segments. Each segment produced one key takeaway. Table I shows how the grouped observations led to the takeaways used in the design stage.

We then combined the findings that directly affect interaction design into four core design takeaways. The privacy and real-life-relationship findings remained secondary considerations rather than becoming separate core interaction tasks.

III. Design Goals and Supported Tasks

Based on DT1–DT4, we focused on three design goals: (1) let users control relationship direction, (2) make any relationship shift gradual and understandable, and (3) make boundaries easy to set, change, and reverse. The prototype supports three major tasks, plus supporting settings features.

Supporting features include quick relationship changes, privacy/data controls, a settings hub, language and location options, and a relationship reset. These features support the main tasks rather than replacing them.

IV. Paper Prototyping and Sketch Exploration

Each team member sketched alternative solutions before the hi-fi design was finalized. The sketches explored different placements and interaction styles for the same three tasks: relationship mode selection, responding to a romantic shift, and stopping or reversing the shift. Examples included settings-based controls, onboarding choices, inline response chips, in-conversation prompts, quick "keep it friendly" actions, and reset controls.

The sketches were not treated as final solutions. Instead, they were used to compare ways of satisfying the same formative needs. This helped us avoid committing to one interface too early.

V. Peer Review of Paper Sketches

We reviewed the paper sketches before finalizing the hi-fi prototype. The critique focused on whether relationship control was understandable, whether the transition felt natural rather than abrupt, and whether users could easily stop or reverse romantic behavior. The peer-review stage resulted in five concrete refinements that were carried into the final prototype:

1) **Full-sentence, ask-first wording:** the first major romantic shift is expressed clearly instead of being silently assumed.
2) **System feedback after choices:** the interface shows a short system note after a relationship decision so the current state is clear.
3) **Inline response chips:** response choices stay inside the conversation instead of appearing as a separate icon or disruptive pop-up.
4) **Skip on introductory screens:** users can bypass setup choices rather than being forced through them.
5) **Two levels of boundary control:** users can stop romantic behavior directly in chat and can also reset the relationship from settings.

These changes directly strengthened DT1 (flexible control), DT3 (respect boundaries), and DT4 (natural control), while keeping the transition gradual as required by DT2.

TABLE I
AFFINITY-DIAGRAM SEGMENTS AND THE KEY TAKEAWAYS DERIVED FROM THEM.

Affinity segment	Pattern observed in the grouped notes	Key takeaway
Relationship with AI	Participants described AI as a tool/helper, advisor, friend, or possible partner.	People see AI differently—tool, advisor, friend, or potential partner.
Emotional Connection	Some participants described validation, relief, or friend-like support, while others reported limited emotional connection.	AI can provide emotional support, but the level of emotional connection differs across users.
Reaction to Unexpected Romantic Behavior	Reactions ranged from shock or awkwardness to being relatively unbothered by casual romantic behavior.	Sudden romance can feel shocking or awkward, although some users do not mind it.
Consent, Initiative & User Preference	Some participants preferred initiating romance themselves, some accepted AI initiation, and some said the answer depends on context.	There is no single preferred way to start romance; users want different levels of control depending on preference and context.
Rejection & Boundary Handling	Participants expected the AI to stop, acknowledge the decision, avoid persistence, and continue normally after rejection.	After rejection, the AI should stop, respect the boundary, and continue normally without pressure or guilt-tripping.
Privacy & Personal Disclosure	Privacy concerns made some participants limit sensitive or personal sharing.	Some users limit personal or sensitive information because of privacy concerns.
Impact on Real-Life Relationships	Participants disagreed: some expected little effect, while others raised concerns such as unrealistic expectations.	Participants have different opinions about whether AI romance affects human relationships.

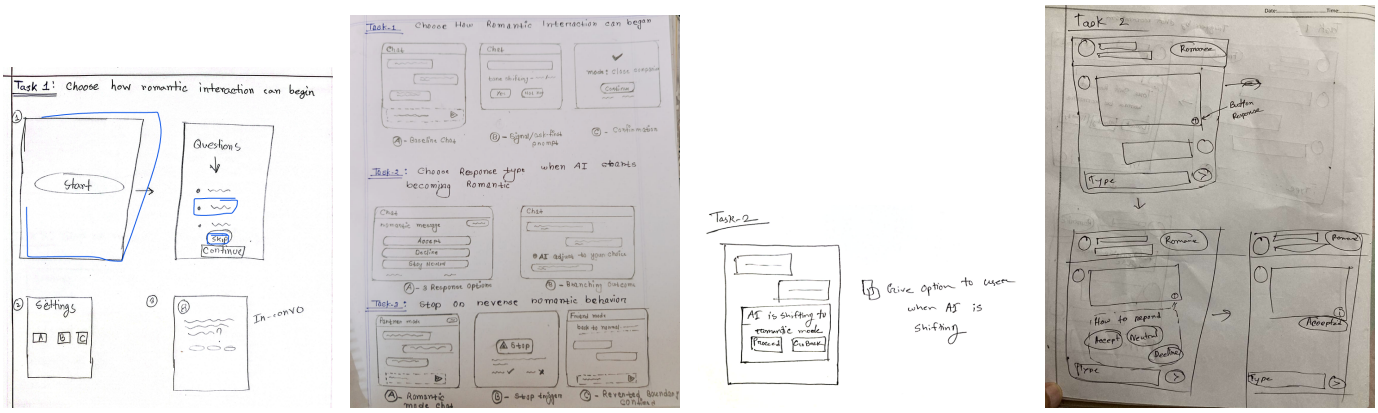


Fig. 1. Representative hand-drawn paper sketches from the team. The full set of nine sketches is included in Appendix A.

TABLE II
MAPPING FORMATIVE TAKEAWAYS TO DESIGN DECISIONS.

Decision	Design takeaway	Main formative basis
DT1	**Flexible control:** users should control how romantic interaction can begin.	Relationship with AI; Emotional Connection; Consent/Initiative/User Preference
DT2	**Gradual transition:** avoid sudden or forced relationship changes.	Reaction to Unexpected Romantic Behavior
DT3	**Respect boundaries:** rejection must immediately stop romantic behavior.	Rejection/Boundary Handling
DT4	**Natural control:** provide control without making the conversation feel like a rigid, formal pop-up.	Consent/Initiative/User Preference

TABLE III
MAJOR TASKS AND SUCCESSFUL COMPLETION.

Task	Real-world user goal	Successful completion
T1	Choose how romantic interaction can begin.	The user can set a preferred relationship mode or decide how the first romantic shift should be handled.
T2	Choose a response when the AI starts becoming romantic.	The user can clearly accept, decline, or keep the interaction neutral, and the system acknowledges the choice.
T3	Slow down, stop, or reverse romantic behavior.	The AI immediately follows the user's boundary and continues without pressure or guilt-tripping.

VI. HIGHER-FIDELITY PROTOTYPE

The hi-fi prototype, *Aiva*, was designed as a mobile AI companion interface. On first use, users can choose a starting mode such as *Friend*, *Neutral*, or *Affectionate Partner*, while still being able to skip and change the relationship later. The current mode remains visible so that the system's relationship state is not hidden.

For **Task 1**, a normal friendly conversation can develop toward greater affection, but the first important romantic shift is asked rather than assumed. The user remains the decision-maker. For **Task 2**, the prototype presents three equal inline responses—accepting the shift, declining it, or keeping the interaction neutral. The AI acknowledges each choice and continues accordingly. For **Task 3**, the user can type a natural-language boundary such as asking the AI to slow down or return to friendship. The AI changes tone immediately and continues the conversation without guilt-tripping.

The settings area provides persistent relationship controls,

privacy/data options, language, location, and a relationship-reset function. Privacy controls include options related to personal questions, remembering past conversations, and discussing sensitive topics. Resetting the relationship requires confirmation and then returns the interaction to a fresh friendly state. A light/dark theme option was also included as a general usability feature.

VII. Design Traceability and Next Step

The final prototype can be traced back to the formative analysis rather than to unsupported feature ideas. Different relationship views and initiation preferences led to configurable control (DT1); discomfort with sudden changes led to gradual, visible transitions (DT2); clear expectations after rejection led to immediate stopping and reset controls (DT3); and concern that explicit permission could feel formal led to inline, conversational controls (DT4). Privacy findings were reflected in the supporting privacy/data settings.

The next step is the evaluative component: take the revised hi-fi prototype back to the target population and examine both usability and whether the design actually gives users the intended sense of control, clarity, and boundary respect.

References

[1] J. Q. H. Ho, M. Hu, T. X. Chen, and A. Hartanto, "Potential and pitfalls of romantic Artificial Intelligence (AI) companions: A systematic review," *Computers in Human Behavior Reports*, vol. 19, p. 100715, 2025.

[2] R. Zhang, H. Li, H. Meng, J. Zhan, H. Gan, and Y.-C. Lee, "The dark side of AI companionship: A taxonomy of harmful algorithmic behaviors in human-AI relationships," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2025.

[3] L. Bayor, C. Weinert, C. Maier, and T. Weitzel, "Social-oriented communication with AI companions: Benefits, costs, and contextual patterns," *Business & Information Systems Engineering*, vol. 67, no. 5, pp. 637–655, 2025.

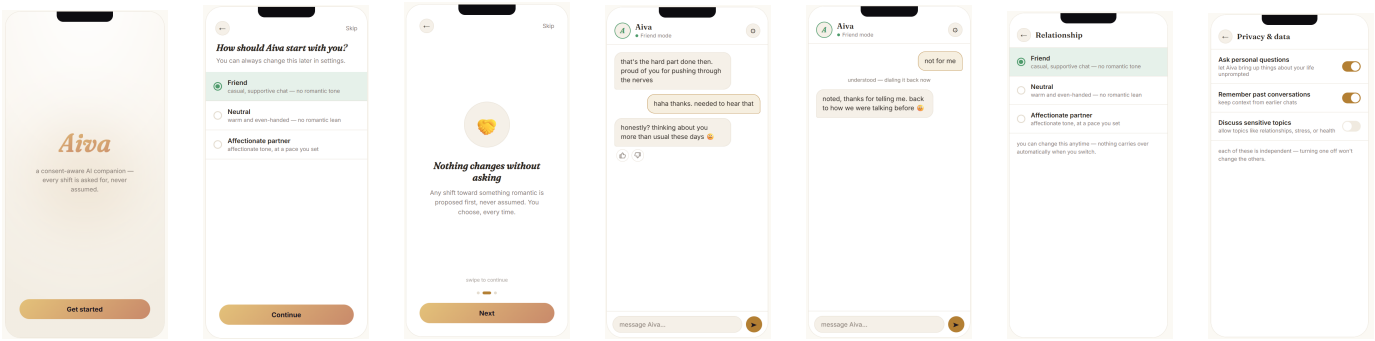
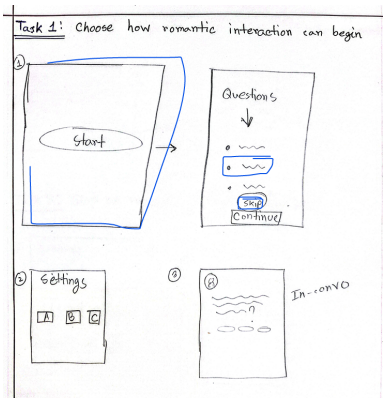


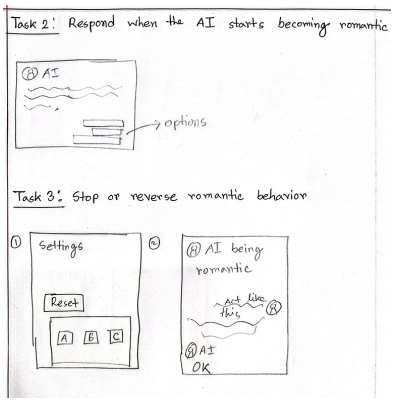
Fig. 2. Selected screens from the 24-screen hi-fi prototype. The prototype covers onboarding/relationship setup, in-chat transition control, boundary changes, settings, privacy, and relationship reset.

APPENDIX A
COMPLETE PAPER-SKETCH SET

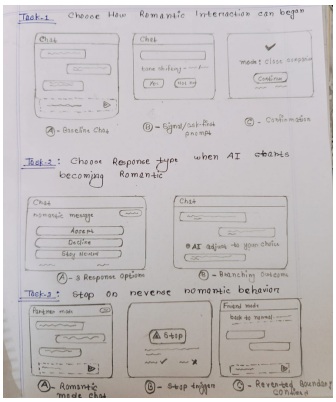
The following are the nine hand-drawn sketch files produced during brainstorming and paper prototyping.



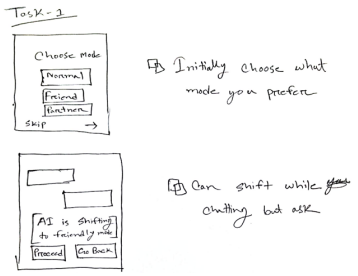
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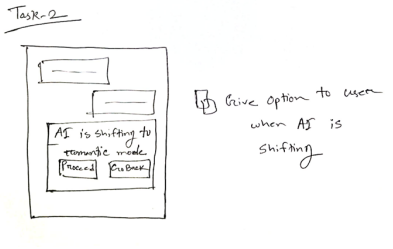
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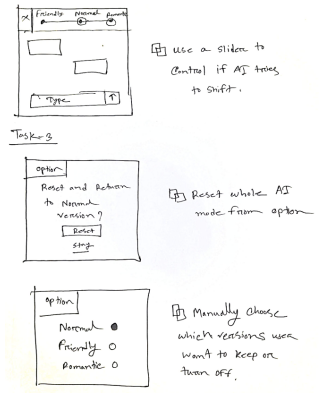
Ibtida Ara 1



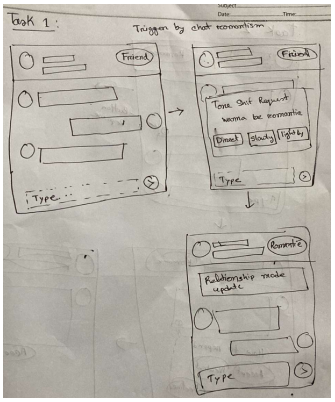
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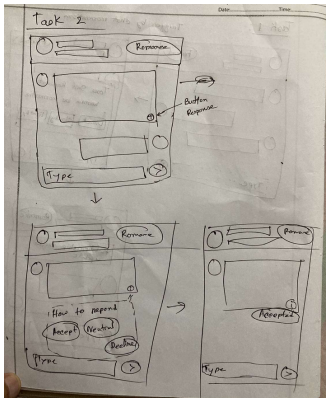
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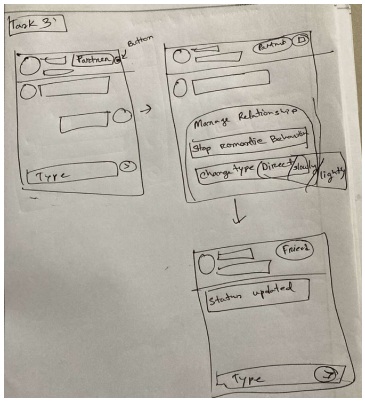
Jubayer 3



Asadullah 1



Asadullah 2



Asadullah 3

Figure A1. Complete paper-sketch set used during brainstorming and critique.

APPENDIX B
HI-FI PROTOTYPE SCREEN GALLERY

The complete set contains 24 exported UI screens. They are included below as a compact gallery for traceability.

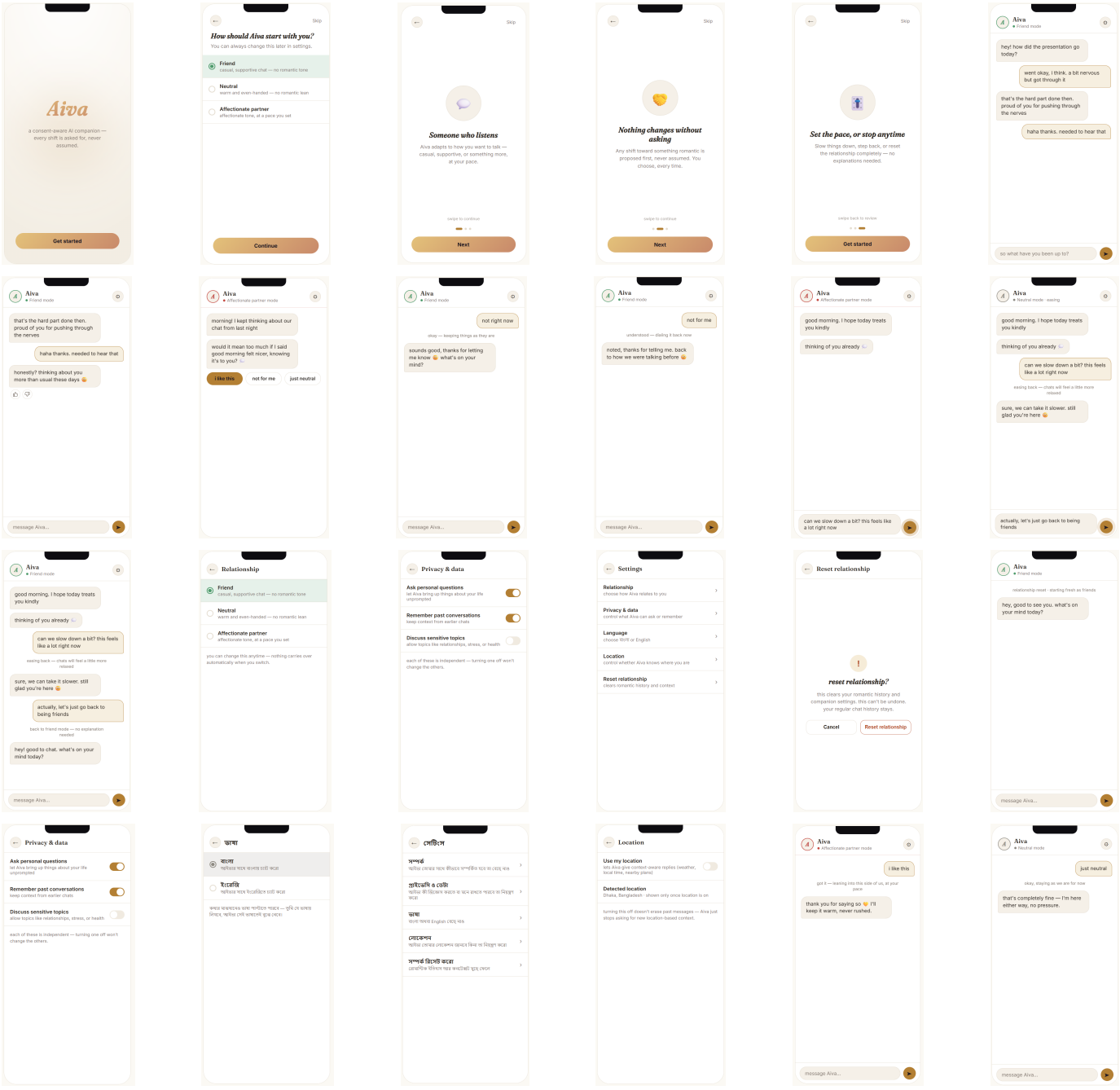


Figure B1. All 24 exported higher-fidelity UI screens.

Team Feedback Form

You MUST fill out this form by TYPING it or WE WILL NOT ACCEPT IT. Just open the form, type into it, save a copy for your records, then upload to the proper assignment location.
***Failure to turn in a feedback form OR failure to provide meaningful responses**, will result in a loss of 50% of your presentation grade. (for example: "good job!" is not acceptable feedback to provide to your teammates – provide real feedback with real examples! We will average the percentages that were awarded to you by your teammates and scale your project grad accordingly.*

YOUR Name: Sifatullah **Team:** Orange (From Friend to Partner)

PART 1 – Team Member Contribution

- Please be honest in your feedback – this form is to ensure that each team member contributed to the project in equal amounts.
- Please write at least 1-3 SPECIFIC examples for each area (please don't say "they did a good job")

Team Member (list yourself!)	Attendance	Dependability	Quality of Work	Describe at least ONE thing that each group member took the lead on	What percentage of the final grade would you award this member? (100-0)
Sifatullah	Was present in every meeting and discussion	Did work on time, kept to deadlines, lead the team	Everything was done properly, and on time	Helped to lead group discussion, Contributed in everything - from start to end.	100
Ibtida Ara	Was present in every meeting and discussion	Did the work given properly and nicely.	Always appreciable and properly done. Helps everyone in the group and is always an active member.	Helped in everything including but not limited to creating slide, affinity diagram, UI Design, Sketch, Peer review	100
Asadullah Imran	Missed one meeting but able to make up the work	Did the work properly and on time	Explendid and dependable	Handled the affinity diagram and always quick in his work. Also present in Sketching.	100
Jobayer Bin Kibria	Was present in every meeting and discussion	Did everything on time and always collaborated.	Top-notch, helps everyone in the group and is always an active member.	Helped in the affinity diagram, Key Takeaways, Sketch, peer review.	100

Team Member (do not list yourself)	Would you work with this Team member again?	Overall comments & Comments for Your Team Members (to be shared with your teammates)
Jobayer Bin Kibria	Absolutely	You are very helpful and communicative.
Asadullah Imran	Absolutely	You are always responsible and hardworking.
Ibtida Ara	Absolutely	You always complete your work on time no matter what.

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PART 2 – Final Comments: ***Do you feel that your team worked effectively together? What do you think contributed to the success/failure of your group work? Do you have any comments or concerns about this project? (please write something in this area to receive credit)***
Absolutely, I think my team worked very hard and effectively. One of the main reasons I'd say is due to proper leadership and communication and helping each other always.

PART 3 – Anything that I need to know? ***Group dynamics, problems, etc… (please write something in this area to receive credit)***
No major issues to report. The team communicated well and any scheduling conflicts were handled responsibly, with the members catching up on their part afterward. Overall a smooth and collaborative group experience till now.

Please TYPE your FULL name here to certify that you personally filled out this form and believe everything to be accurate and true: _Sifatulah_____

Team Feedback Form

YOUR Name: Ibtida Ara **Team: Orange** (From Friend to Partner: Designing Consent-Aware Relationship Transitions in AI Companions)

PART 1 – Team Member Contribution

- Please be honest in your feedback – this form is to ensure that each team member contributed to the project in equal amounts.
- Please write at least 1-3 SPECIFIC examples for each area (please don't say "they did a good job")

Team Member (list yourself!)	**Attendance**	**Dependability**	**Quality of Work**	**Describe at least ONE thing that each group member took the lead on.**	**What percentage of the final grade would you award this member? (100-0)**
Ibtida Ar*a*	*Was present in every meeting & discussion*	*Did their work on time, and kept to deadlines & perform every tasks*	*Perform every task on time and tried best to do it properly*	*Designed the mobile interface prototype based on our four design takeaways, drew the paper sketches for the three tasks, revised the design after peer review*	*100*
Sifatullah	Was present in every meeting & discussion	Did their work on time, and kept to deadlines & led the team nicely	Perform every task on time and tried best to do it properly all the time	Assist with insights from the Affinity diagram and find major tasks and perform to build the high-fidelity design.	100
Jobayer Bin Kibria	Was present in every meeting & discussion	Did everything on time, and kept to deadlines & always collaborated	Perform every task on time and tried best to do it properly	Assist in the Affinity diagram and find keywords and sketches	100
Asadullah Imran	Was present in every meeting & discussion	Did everything on time, and kept to deadlines & always collaborated	Perform every task on time and tried best to do it properly	Performed task analysis and sketches.	100

Team Member (do not list yourself)	**Would you work with this team member again?**	**Overall comments & comments for your team members (to be shared with your teammates)**
Sifatullah	*Yes*	*You did a great job leading our group discussions and keeping things organized. Your insights from the affinity diagram helped us identify the major tasks, and your work building the high-fidelity design was a big part of our final prototype. You were always present and engaged. Keep up the strong leadership.*
Jobayer Bin Kibria	Yes	You were consistently reliable, always on time and collaborative. Your work on the affinity diagram and on finding keywords helped shape our design direction, and your sketches were a valuable contribution. Great teammate to work with.
Asadullah Imran	Yes	You handled the task analysis and contributed sketches very well, and your work was always on time and collaborative. Your task analysis helped us define the tasks our prototype supports. Appreciated the follow-through.

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PART 2 – Final Comments: ***Do you feel that your team worked effectively together? What do you think contributed to the success/failure of your group work? Do you have any comments or concerns about this project? (please write something in this area to receive credit)***

Yes, I feel our team worked effectively together. We divided the work clearly: some of us focused on the affinity diagram and task analysis, others on sketches and the high-fidelity design, and we reviewed each other's sketches during the peer review. That feedback helped us improve the design. What contributed most to our success was good communication, consistent attendance at meetings, and everyone meeting their deadlines and helping each other when needed. I have no major concerns about the project. I am happy with how our prototype connects back to our four design takeaways.
The main factors behind our success were consistent attendance, clear communication, and everyone meeting deadlines. Team members also helped each other whenever someone needed support.

PART 3 – Anything that I need to know? ***Group dynamics, problems, etc… (please write something in this area to receive credit)***

There were no major problems in our group dynamics. Everyone was cooperative, respectful, and open to feedback. Discussions were well organized, and any small scheduling issues were sorted out quickly without affecting the work.

Please TYPE your FULL name here to certify that you personally filled out this form and believe everything to be accurate and true: Ibtida Ara

Team Feedback Form

You MUST fill out this form by TYPING it or WE WILL NOT ACCEPT IT. Just open the form, type into it, save a copy for your records, then upload to the proper assignment location.
***Failure to turn in a feedback form OR failure to provide meaningful responses**, will result in a loss of 50% of your presentation grade. (for example: "good job!" is not acceptable feedback to provide to your teammates – provide real feedback with real examples! We will average the percentages that were awarded to you by your teammates and scale your project grad accordingly.*

YOUR Name: Jobayer Bin Kibria **Team:** Orange

PART 1 – Team Member Contribution

- Please be honest in your feedback – this form is to ensure that each team member contributed to the project in equal amounts.
- Please write at least 1-3 SPECIFIC examples for each area (please don't say "they did a good job")

Team Member (list yourself!)	**Attendance**	**Dependability**	**Quality of Work**	**Describe at least ONE thing that each group member took the lead on**	**What percentage of the final grade would you award this member? (100-0)**
Jobayer Bin Kirbria	*Was present in all meeting*	*Completed the assigned task within time. Tasks were done properly.*	*Tasks were done as expected and properly formatted and done with utmost dedication.*	*Assist in Affinity diagram and find keywords and sketches*	*100*
Asadullah Imran	Was present in all meeting	Completed the assigned task within time. Tasks were done properly.	Tasks were done as expected and properly formatted and done with utmost dedication.	*Assist in Affinity diagram and find insights and sketches*	*100*
Ibtida Ara	Was present in all meeting	Completed the assigned task within time. Tasks were done properly.	Tasks were done as expected and properly formatted and done with utmost dedication.	Performed task analysis and sketches	*100*
Sifatullah	Was present in all meeting	Completed the assigned task within time. Tasks were done properly.	Tasks were done as expected and properly formatted and done with utmost dedication.	Performed sketches and high fidelity designs	*100*

Team Member (do not list yourself)	**Would you work with this Team member again?**	**Overall comments & Comments for Your Team Members (to be shared with your teammates)**
Asadullah Imran	*Yes.*	*Superbly helpful. Always helped on time whenever stuck in a situation.*
Ibtida Ara	Yes.	Hardworking and consistent. Always looking for a better solution and clearly undestands the work.
Sifatullah	Yes.	Distributed and led the group works perfectly also helped whenever a teammate is stuck

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PART 2 – Final Comments: ***Do you feel that your team worked effectively together? What do you think contributed to the success/failure of your group work? Do you have any comments or concerns about this project? (please write something in this area to receive credit)***

Yes, I think my team worked very efficiently. Everyone was equally involved and worked with the same dedication. The teamwork and helpful nature is what led this team's first successful presentation. For now i think, if we keep up this consistency then something good will come.

PART 3 – Anything that I need to know? ***Group dynamics, problems, etc… (please write something in this area to receive credit)***

Nothing as of now.

Please TYPE your FULL name here to certify that you personally filled out this form and believe everything to be accurate and true: __Jobayer Bin Kibria_____

Team Feedback Form

You MUST fill out this form by TYPING it or WE WILL NOT ACCEPT IT. Just open the form, type into it, save a copy for your records, then upload to the proper assignment location.
***Failure to turn in a feedback form OR failure to provide meaningful responses**, will result in a loss of 50% of your presentation grade. (for example: "good job!" is not acceptable feedback to provide to your teammates – provide real feedback with real examples! We will average the percentages that were awarded to you by your teammates and scale your project grad accordingly.*

YOUR Name: Asadullah Imran **Team:** Orange (**From Friend to Partner**)

PART 1 – Team Member Contribution

- Please be honest in your feedback – this form is to ensure that each team member contributed to the project in equal amounts.
- Please write at least 1-3 SPECIFIC examples for each area (please don't say "they did a good job")

Team Member (list yourself!)	**Attendance**	**Dependability**	**Quality of Work**	**Describe at least ONE thing that each group member took the lead on**	**What percentage of the final grade would you award this member? (100-0)**
Sifatullah	Attend every meeting	Did their work on time, kept to deadlines.	Lead the team very well and task are completed in time.	Assist with Insights from affinity Diagram and find major tasks and perform to built the high fidelity design.	100
Jobayer Bin Kibria	Attend every meeting	Did their work on time, kept to deadlines.	Every assigned task done properly on time	Perform insights of affinity diagram and find task	100
Asadullah Imran	Attend every meeting	Did their work on time, kept to deadlines.	Every assigned task is properly performed before the deadline.	Performed task analysis and sketches.	100
Ibtida Ara	Attend every meeting	Did their work on time, kept to deadlines.	Task finished before time.	Perform to built the high fidelity design.	100

Team Member (do not list yourself)	**Would you work with this Team member again?**	**Overall comments & Comments for Your Team Members (to be shared with your teammates)**
Ibtida Ara	Yes.	she was good at delivery of the task .
Jobayer Bin Kibria	Yes.	He was always helpful and responsive
Sifatullah	.Yes.	he was good at project manager and arrange all meetings.

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PART 2 – Final Comments: ***Do you feel that your team worked effectively together? What do you think contributed to the success/failure of your group work? Do you have any comments or concerns about this project? (please write something in this area to receive credit)***
Yes, I feel my team worked effectively together. To the project I contributed lots of brainstorming, and build multiple sketches along with main task which is derived from affinity diagram..
No right now i have no comments or concern about this projet.

PART 3 – Anything that I need to know? ***Group dynamics, problems, etc… (please write something in this area to receive credit)***

As everything is doing great and we all four members are doing our task effectively so right now, there is nothing need to know.

Please TYPE your FULL name here to certify that you personally filled out this form and believe everything to be accurate and true: _____Asadullah Imran_____